

Training Theory: Stability, Calibration, Compression, and Distillation

Mathematics of Deep Learning — Chapter 14

Yin Yang

Foundation Books Project

Roadmap

Training Loss vs Deployment Behavior

Stability as Sensitivity

Sharpness, Flatness, Local Robustness

Regularization Revisited

Calibration and Confidence

Compression as Approximation

Knowledge Distillation

Fine-Tuning, Forgetting, Drift

Training Diagnostics Decision Guide

Summary

Training Loss vs Deployment Behavior

After the Run Finished

Earlier chapters built losses, optimizers, and architectures. This chapter asks what happens *after* training.

- Running example: 3-class classifier with logits $g_{\theta}(x) \in \mathbb{R}^3$ and $p_{\theta}(y | x) = \text{softmax}(g_{\theta}(x))$.
- Sample value: logits $(2, 1, 0) \Rightarrow$ probabilities $(0.665, 0.245, 0.090)$.
- Four classes of check: *generalization, stability, calibration, compression/distillation*.

A trained model still has to prove it behaves well outside the training loop.

Four Risks, Four Roles

Training, validation, test empirical risks

$$\hat{R}_{\bullet}(\theta) = \frac{1}{|S_{\bullet}|} \sum_{(x,y) \in S_{\bullet}} \ell(f_{\theta}(x), y), \quad \bullet \in \{\text{train, val, test}\}.$$

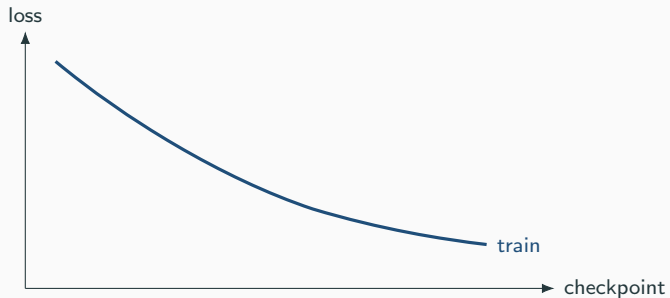
Deployment risk

$$R_{\text{deploy}}(\theta) = \mathbb{E}_{(X,Y) \sim \mathcal{D}_{\text{deploy}}} [\ell(f_{\theta}(X), Y)].$$

Validation generalization gap: $G_{\text{val}}(\theta) = \hat{R}_{\text{val}}(\theta) - \hat{R}_{\text{train}}(\theta).$

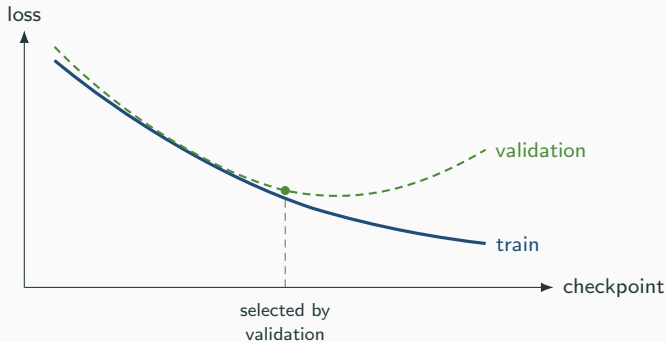
Validation is one held-out distribution; deployment may be another.

Loss Curves: Train, Val, Deploy



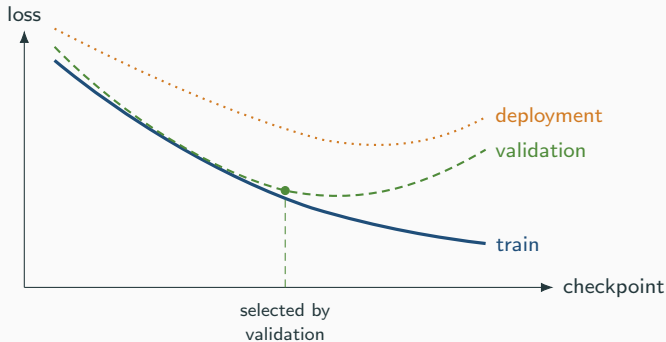
(1) Training loss can keep decreasing...

Loss Curves: Train, Val, Deploy



- (1) Training loss can keep decreasing. . .
- (2) . . . while validation has already stopped improving.

Loss Curves: Train, Val, Deploy



- (1) Training loss can keep decreasing. . .
- (2) . . . while validation has already stopped improving.
- (3) Deployment curve sits elsewhere: different distribution.

Train, val, deploy are different mathematical roles.

Validation Concentration (Hoeffding)

Fix θ *before* validation, draw n iid val examples with loss in $[0, B]$. Then

$$\mathbb{P}\left(\left|\hat{R}_{\text{val}}(\theta) - R_{\text{val}}(\theta)\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{2n\epsilon^2}{B^2}\right).$$

Equivalently, with probability $\geq 1 - \delta$,

$$\left|\hat{R}_{\text{val}}(\theta) - R_{\text{val}}(\theta)\right| \leq B \sqrt{\frac{\log(2/\delta)}{2n}}.$$

- Requires θ fixed *before* inspecting val data.
- Repeated model selection on the same val set weakens the bound.
- Concentration around R_{val} does not control R_{deploy} if $\mathcal{D}_{\text{deploy}} \neq \mathcal{D}_{\text{val}}$.

The bound estimates one distribution, not every future use.

Stability as Sensitivity

Learning Algorithm A Maps Data to Parameters

A learning algorithm A takes a training set S and returns fitted parameters $\theta_S = A(S)$.

Leave-one-out loss sensitivity at validation point (x, y) :

$$\Delta_i(x, y) = \left| \ell(f_{A(S)}(x), y) - \ell(f_{A(S \setminus i)}(x), y) \right|.$$

Example. Prediction $(0.66, 0.25, 0.09)$ becomes $(0.61, 0.28, 0.11)$ after removing one training example. Max change: 0.05.

- Compares two fitted models on the *same* val point.
- Asks how much removing one example moves the predictor.

Stability asks how much the fitted predictor changes after a small training change.

Uniform Stability Bounds the Generalization Gap

β -uniform stability

For bounded loss, every S , removed index i , and checked (x, y) :

$$|\ell(f_{A(S)}(x), y) - \ell(f_{A(S \setminus i)}(x), y)| \leq \beta.$$

Expected gap bound

$$|\mathbb{E}_S[R(A(S)) - \hat{R}_S(A(S))]| \leq \beta.$$

Stability levers (reduce β in practice): smaller learning rate, early stopping, larger val sets, weight decay, less aggressive fine-tuning.

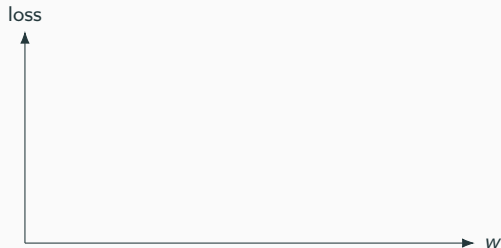
Many practical training choices are stability choices in disguise.

Sharpness, Flatness, Local Robustness

Sharpness Proxy and Two Minima

Local perturbation set: $B_\rho(\theta) = \{\theta + \delta : \|\delta\|_2 \leq \rho\}$.

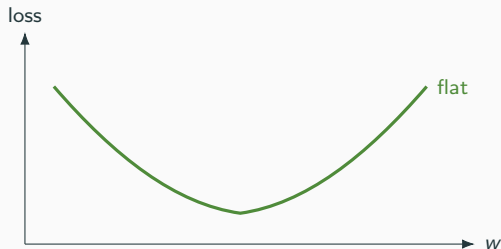
$$\text{sharp}_\rho(\theta) = \max_{\|\delta\|_2 \leq \rho} (L(\theta + \delta) - L(\theta)).$$



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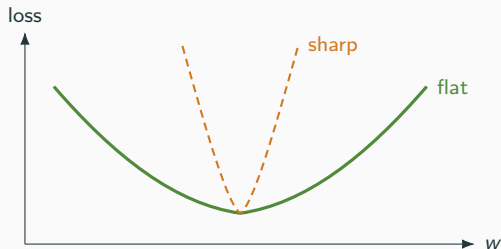


Flat $(w - 1)^2$: $\delta = 0.1$ raises loss by 0.01.

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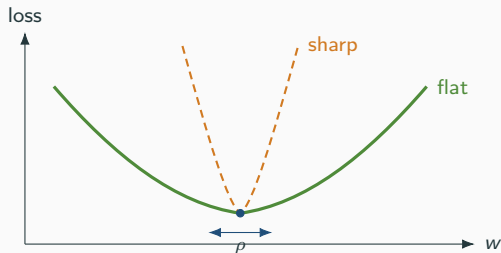
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Sharp $20(w - 1)^2$: same δ raises loss by 0.20.

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Flat $(w - 1)^2$: $\delta = 0.1$ raises loss by 0.01.

Sharp $20(w - 1)^2$: same δ raises loss by 0.20.

Same minimum value, different local sensitivity.

Curvature, Hessian, and Caveats

Quadratic model near a stationary point:

$$L(\theta + \delta) \approx L(\theta) + \nabla L(\theta)^\top \delta + \frac{1}{2} \delta^\top H(\theta) \delta.$$

For 1D $L(w) = L(w_0) + a(w - w_0)^2$, $\max_{|\delta| \leq \rho} \Delta L = a\rho^2$. Doubling ρ (e.g. quantization) multiplies the worst-case rise by 4.

- Sharpness depends on parameterization and scale.
- Compare proxies only under same architecture, scale, and eval.
- Useful for explaining quantization / fine-tuning sensitivity.

Sharpness is a local diagnostic, not a complete generalization theory.

Regularization Revisited

Regularizers: Name What They Change

$$J(\theta) = \hat{R}_{\text{train}}(\theta) + \lambda \Omega(\theta).$$

Method	Object changed	Purpose
Weight decay	objective ($\lambda \ \theta\ _2^2$)	discourage large weights
Dropout	training-time computation	reduce co-adaptation
Label smoothing	target distribution	avoid one-hot certainty
Mixup	examples and labels	smooth behavior between examples
Early stopping	chosen checkpoint	stop before val degrades

Label-smoothed target for $K = 3$, $\epsilon = 0.15$: $(1, 0, 0) \mapsto (0.90, 0.05, 0.05)$.

A regularizer should be named by what it changes.

Implicit max-margin (informal)

For separable logistic regression, GD drives loss toward zero and $w_t / \|w_t\|_2 \rightarrow w_{\text{SVM}} / \|w_{\text{SVM}}\|_2$.

PAC-Bayes (informal)

With probability $\geq 1 - \delta$, posterior Q over predictors:

$$\text{kl}(\widehat{L}_S(Q) \parallel L(Q)) \leq \frac{D_{\text{KL}}(Q \parallel P) + \log((n+1)/\delta)}{n}.$$

Benign overfitting: zero training error can still predict well in some regimes. NTK: a fixed-kernel reference for wide-network dynamics.

The optimizer chooses among many zero-error solutions; data geometry decides if interpolation is benign.

Calibration and Confidence

Confidence vs Correctness

Top-class confidence

$$\hat{y}(x) = \arg \max_k p_{\theta}(y = k \mid x), \quad C(x) = \max_k p_{\theta}(y = k \mid x).$$

Calibration (population): $\mathbb{P}(Z = 1 \mid C = c) = c$.

Among predictions with confidence c , fraction correct should be c .

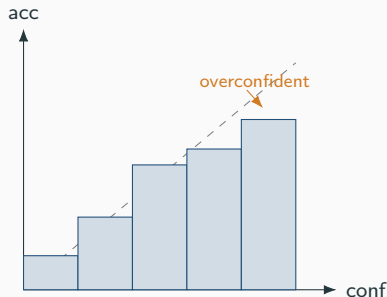
- Accuracy: did we pick the right class?
- Calibration: does $C(x)$ have the right frequency meaning?
- A model can be accurate and overconfident.

Calibration is about probability *meaning*, not just class accuracy.

Expected Calibration Error and Reliability Diagram

Bin predictions $B_1, \dots, B_M$ over N examples:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|.$$



Caveats: ECE depends on bins, binning rule, and N . Empty bins are noisy.

Report ECE *with* its binning rule and sample size.

Temperature Scaling Preserves Ranking

Post-training temperature $\tau > 0$:

$$\hat{\tau} = \arg \min_{\tau > 0} -\frac{1}{n} \sum_{i=1}^n \log \frac{\exp(z_{i,y_i}/\tau)}{\sum_k \exp(z_{i,k}/\tau)}.$$

Logits (2, 1, 0):

τ	probabilities	reading
0.5	(0.867, 0.117, 0.016)	sharper
1.0	(0.665, 0.245, 0.090)	original
2.0	(0.506, 0.307, 0.186)	softer

Order is preserved $\Rightarrow$ top-1 accuracy unchanged. Temperature scaling cannot repair misranked classes.

Overconfident on val $\Rightarrow$ raise τ ; underconfident $\Rightarrow$ lower τ .

Compression as Approximation

Compression: Approximation with a Resource Budget

Replace f_θ by cheaper $\tilde{f}$. Prediction-space error:

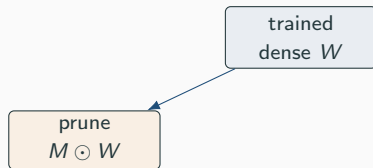
$$E_{\text{comp}} = \frac{1}{|S_{\text{val}}|} \sum_{x \in S_{\text{val}}} \|f_\theta(x) - \tilde{f}(x)\|_1.$$

trained
dense W

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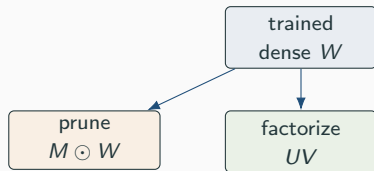
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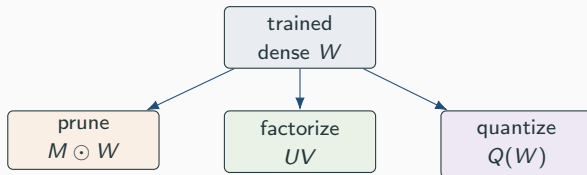
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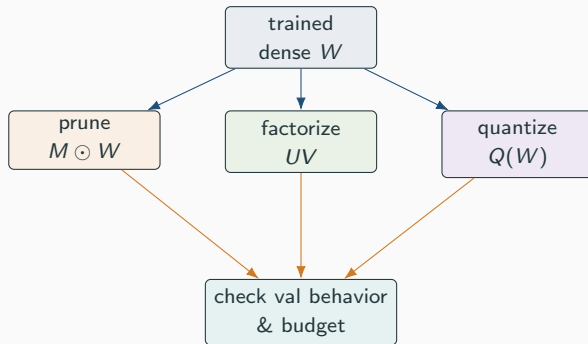
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Compression error is rarely uniform across slices.

Eckart–Young, Low-Rank, Quantization

Eckart–Young–Mirsky

For $W = U\Sigma V^\top$, the rank- r truncated SVD W_r minimizes $\|W - A\|_F$ (and spectral norm) over $\text{rank}(A) \leq r$.

Dense $m \times n$ uses mn weights; rank- r uses $r(m + n)$. Saves weights iff $r < mn/(m + n)$.

Example: $m = 1000$, $n = 500$, $r = 50$: 500 000 $\rightarrow$ 75 000 weights (15%).

Uniform scalar quantization

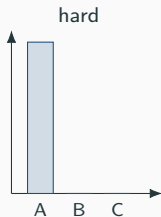
$Q(w) = s \cdot \text{round}(w/s)$, step $s > 0$.

Best-approximation for one matrix is not best-approximation for one task.

Knowledge Distillation

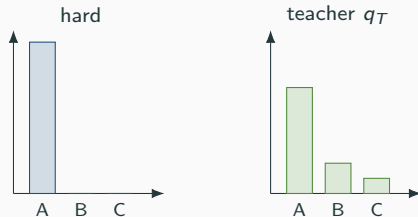
Soft Targets Carry More Than the Winner

Hard label $e_A = (1, 0, 0)$ vs teacher $q_T = (0.70, 0.20, 0.10)$.



Soft Targets Carry More Than the Winner

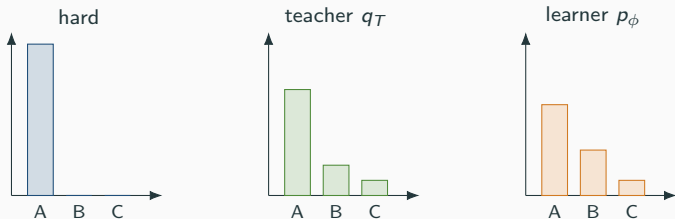
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(2) Teacher: B more plausible than C — extra signal.

Soft Targets Carry More Than the Winner

Hard label $e_A = (1, 0, 0)$ vs teacher $q_T = (0.70, 0.20, 0.10)$.



- (2) Teacher: B more plausible than C — extra signal.
- (3) Learner trained to match the teacher's full distribution.

A soft teacher target says more than the winning class.

Distillation KL and Combined Objective

$$L_{\text{distill}}(\phi; x) = D_{\text{KL}}(q_{\mathcal{T}}(\cdot | x) \| p_{\phi}(\cdot | x)) = \sum_k q_{\mathcal{T}}(k | x) \log \frac{q_{\mathcal{T}}(k | x)}{p_{\phi}(k | x)}.$$

Gibbs: $D_{\text{KL}} \geq 0$ with equality iff $p_{\phi} = q_{\mathcal{T}}$.

Combined objective

$$L_{\text{learn}} = (1 - \alpha) \text{CE}(e_y, p_{\phi}) + \alpha D_{\text{KL}}(q_{\mathcal{T}} \| p_{\phi}).$$

Learner $p = (0.60, 0.30, 0.10)$, hard label A : $-\log(0.60) \approx 0.511$. Teacher loss also checks B and C . Temperature on the teacher exposes non-winner similarities.

Tune α : how much teacher behavior vs. how much hard labels.

Fine-Tuning, Forgetting, Drift

Parameter Drift and Retained-Slice Regression

Parameter drift

$$D_{\text{param}} = \|\theta_1 - \theta_0\|_2.$$

Retained-slice regression

$$\Delta_{\text{retain}} = \hat{R}_{S_{\text{retain}}}(\theta_1) - \hat{R}_{S_{\text{retain}}}(\theta_0).$$

Slice	before	after	reading
Target task accuracy	71%	84%	improved
Retained source acc.	88%	79%	forgetting
Calibration ECE	0.04	0.11	worse confidence

Improving a target task can damage retained behavior and calibration.

Anchor Regularizer: Penalize Drift

$$J(\theta) = \hat{R}_{\text{target}}(\theta) + \lambda \|\theta - \theta_0\|_2^2, \quad \nabla J = 2\lambda(\theta - \theta_0).$$

- Weight decay pulls θ toward 0.
- Anchor penalty pulls θ toward the *pretrained* θ_0 .
- Related to continual-learning methods that reduce catastrophic forgetting.

Fine-tuning reports should record: starting checkpoint, trainable layers, data size, seeds, validation slices, retained-behavior checks.

Name the retained slice and metric *before* declaring fine-tuning helped.

Training Diagnostics Decision Guide

Symptom → Measurement, Step by Step

symptom
(visible)

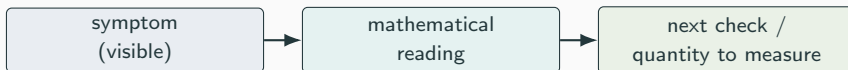
(1) Start from the visible symptom.

Symptom → Measurement, Step by Step



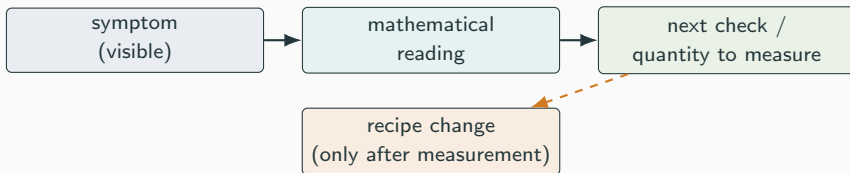
- (1) Start from the visible symptom.
- (2) Translate to a mathematical quantity.

Symptom → Measurement, Step by Step



- (1) Start from the visible symptom.
- (2) Translate to a mathematical quantity.
- (3) Pick the measurement that decides between causes.

Symptom → Measurement, Step by Step



- (1) Start from the visible symptom.
- (2) Translate to a mathematical quantity.
- (3) Pick the measurement that decides between causes.
- (4) Only then change the training recipe.

Name the measured quantity before changing the recipe.

The Diagnostic Table

Symptom	Math reading	Next check
Train falls, val rises	positive gen. gap	splits, capacity, reg., early stop
High acc, high ECE	over/underconfident	reliability table, temperature
Seeds vary widely	training is unstable	report spread; reduce LR
Avg acc kept, rare class drops	uneven approx. error	per-slice metrics
Top-1 matches, conf. poor	probabilities not transferred	tune α , calibrate
Target up, retained down	drift / forgetting	retained slice, anchor

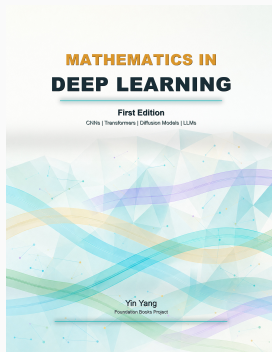
A model card should record: split, checkpoint, train+val loss, slice metrics, calibration summary, model size, compression / distillation objective.

The measurement identifies the problem being solved.

Summary

- **Deployment $\neq$ validation:** four risks, four roles; Hoeffding only handles fixed θ on one distribution.
- **Stability:** how much the fitted predictor moves when data or randomness moves; bounds the expected gen. gap.
- **Sharpness:** local sensitivity to parameter perturbations; depends on scale, useful as a relative diagnostic.
- **Regularization:** name the object changed — objective, target, data, computation, or checkpoint.
- **Calibration:** probability meaning, measured with ECE and adjusted with temperature; ranking unchanged.
- **Compression & distillation:** approximate the model under a budget; transfer the full distribution, not just the top class.
- **Fine-tuning:** target improvement does not imply retained improvement; anchor and slice metrics make drift visible.

Next: scaling laws, evaluation under shift, and decision-quality metrics for deployed models.



Mathematics in Deep Learning

Chapter overview slides for the book by Yin Yang.

Book page	foundation-books.github.io/mathematics-in-deep-learning
Code	github.com/foundation-books/mathematics-in-deep-learning-code
Paperback	amazon.com/dp/BOH1ZZHDT2
Hardcover	amazon.com/dp/BOH1ZL11MJ
Kindle	amazon.com/dp/B0GX32M8SP
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